Morse–Thue Sequence (9.35)

A. Engel — Problem-Solving Strategies · 1998

Morse–Thue Sequence. Start with 0 ; to each initial segment append its complement : 0, 01, 0110, 01101001, . . .

(a) Let the digits of the sequence be $x(0), x(1), x(2), \ldots$. Prove that $x(2n) = x(n)$, $x(2n+1) = 1 - x(2n)$.

(b) Prove that $x(n) = 1 - x(n - 2^k)$, where 2^k is the largest power of 2 which is $\leq n$. Find the 1993rd digit of the sequence.

(c) Prove that the sequence is not periodic.

(d) Write the nonnegative integers in base 2 : $0, 1, 10, 11, \ldots$. Now replace each number by the sum of its digits mod 2. You get the Morse–Thue sequence. Prove this.